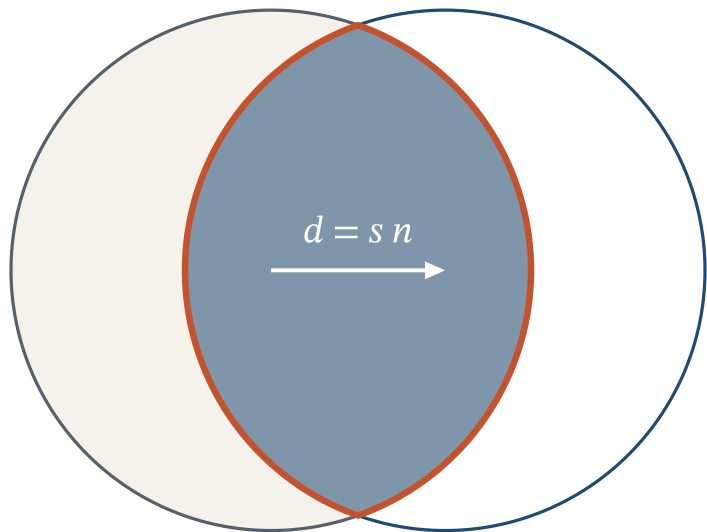


Incident ball $|y| \leq B$



Observation ball $|y - d| \leq B$

Dark lens: retained domain; vermilion boundary: sphere supporting both fluxes

Even a constant kernel has a cusp

$$\text{Vol}(\text{lens}) = 4\pi B^3/3 - \pi B^2|s| + \pi|s|^3/12$$

$$0 \leq |s| \leq 2B$$

Quadratic coefficient $\neq$ origin Hessian of the full cusp response

Fixed input; expand the exact radial endpoint

$$\text{body: } \frac{1}{2} \int_{\text{ball}} \text{Re } R_{\text{dd}}$$

$$\text{first flux: } \frac{1}{2} \int_{\text{sphere}} u \text{Re } R_{\text{d}}$$

$$\text{zeroth flux: } \frac{1}{4} \int_{\text{sphere}} [u^2 \partial_r \text{Re } R_0 + (3u^2 - 1) \text{Re } R_0/B]$$

$$\text{Affine-field check: } a(k) = [1 + (2+i)k_3] b, b^T Q b = 4$$

Boundary flux = nonzero gradient energy $\rightarrow$ quadratic coefficient 0

Deleting the flux creates a spurious negative quadratic term. Volume and area retain the original $(2\pi)^{-3}$ measure.